Rapid Prototyping 161

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Class Outline

Introduction to Additive Manufacturing
Design for Additive Manufacturing
Prototyping
Rapid Prototyping Types
Design and Prototyping Review
Rapid Processes for Prototyping
Rapid Prototyping Applications
Rapid Prototyping: Next Steps
Final Review

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Introduction to Additive Manufacturing

Additive manufacturing (AM) is a newer manufacturing process that builds three-dimensional (3D) objects by adding one layer of material at a time. There are a variety of AM methods that use a number of different materials such as plastic or metal. In contrast, most traditional manufacturing processes are subtractive, as they remove, or subtract, pieces of a material in order to create an object or part.

Today, AM methods, which may be referred to as 3D printing, have many applications in manufacturing including the creation of functional end-use products. However, AM began in the 1980s with rapid prototyping, a process of quickly creating prototypes to test designs and support new products in traditional manufacturing. Since those early days, rapid prototyping techniques have expanded to include a wide variety of AM methods and materials. These AM techniques enable manufacturers to create lighter and more complex AM prototypes and end-use products without sacrificing quality.

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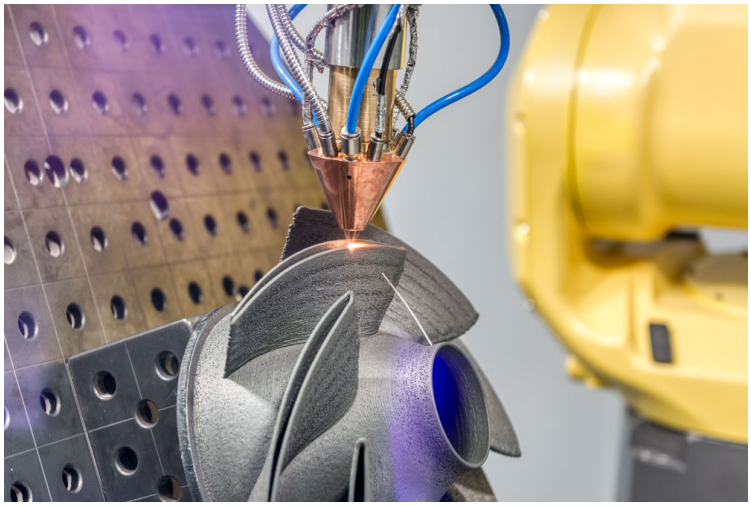

Figure 1: Additive technology for sintering metal parts.

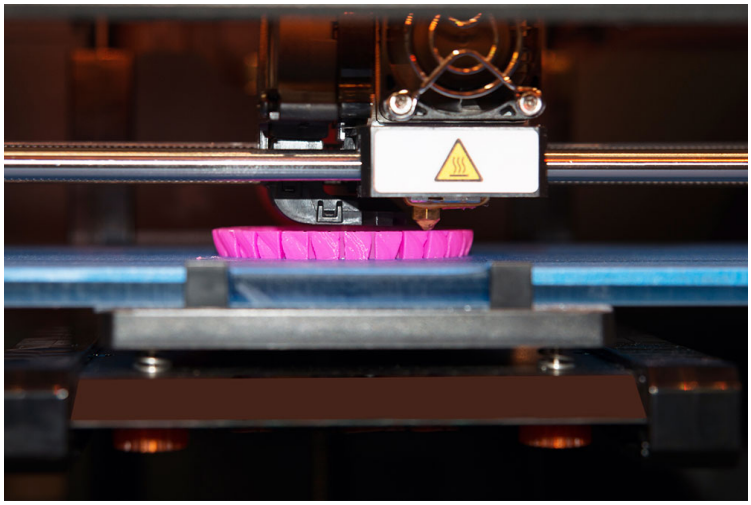

Figure 2: Fused deposition modeling used to make a three-dimensional object.

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Lesson: 2/9

Design for Additive Manufacturing

Conceptualization refers to the formation of an idea about a possible part or product. It represents the initial design stage for manufacturing.

The planning stage involves figuring out the best way to create the initial design, or concept. Planning considers how an innovative concept might fit together with the practical concerns related to manufacturing processes.

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The designing stage begins with the process of creating the actual specifications for a prototype or potential part. Developing a prototype can include such things as creating a blueprint or digital model with computer-aided design (CAD). Rapid prototyping is the process of using AM operations to create prototypes for parts that will then be produced through traditional manufacturing.

Ultrasonic testing prototype.

The testing stage examines a part prototype to ensure that it performs its designed function when manufactured. Testing can indicate if a part is ready for production or needs additional planning and designing.

The production stage represents the final stage of design in manufacturing design. The production process creates the tested design and delivers it to a customer as specified.

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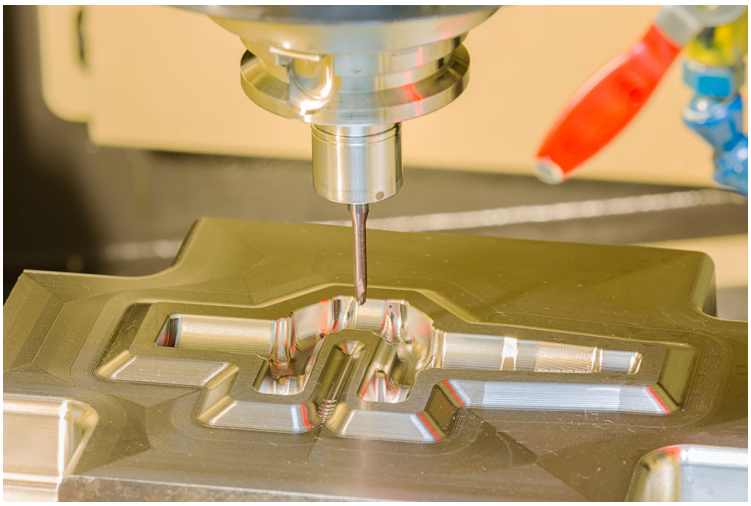
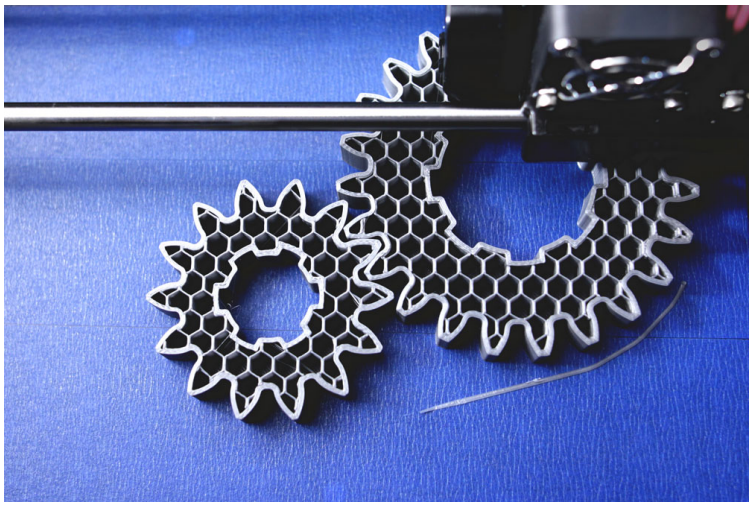
Prototyping

Planning, designing, and testing represent the steps most associated with the development of a prototype, the preliminary model of a part used to evaluate the performance and functionality of a design. Effective prototyping ensures the manufacturability of the part during the production stage.



CNC mill creating a grooved surface.

Prototyping for traditional manufacturing has limitations based on the raw materials available. For example, bar stock can come in round, rectangular, and square set forms which have limits as to how tools can reshape the material. The tools used to shape the materials also have limits. For example, a computer numerical control (CNC) mill can create excellent flat or grooved surfaces, but has difficulty creating more complex part shapes. Using subtractive manufacturing processes like CNC machining for prototyping also requires a different machine and tool setup for each prototype, which is time consuming and expensive.



Complex topology of an AM part.

Unlike traditional manufacturing, AM prototyping does not require machine reconfiguration and retooling, which can greatly shorten the planning, designing, and testing steps. This enables manufacturers to focus on the capabilities and complexities of a prototype. For example, AM prototypes can create lighter, stronger, and more complex parts for aircraft, which leads to better performance and lower operating costs. Although AM offers more flexibility with materials, AM machines can typically process only one category of material at a time. For example, direct metal laser sintering (DMLS) can only create metal objects. In addition, unique challenges associated with AM prototypes include the use of a support structure and issues with topology, or the ways in which geometric or material shapes get arranged in a part.

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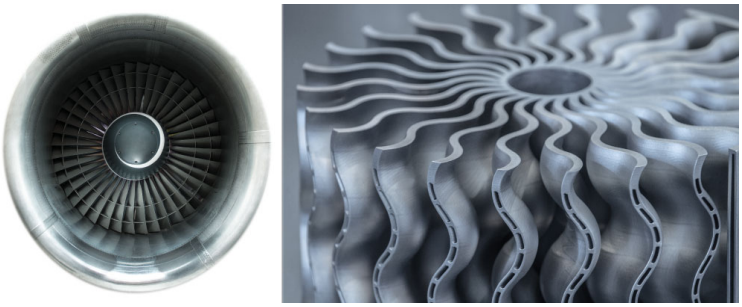
Lesson: 4/9

Rapid Prototyping Types



AM rapid prototypes are often conceptual models, which are non-functional part prototypes that allow designers to provide a physical example of their design ideas. Conceptual models may be made at a smaller size or out of a less expensive material than a functional part in order to save time and costs, especially if the design is likely to change and require additional prototypes. For instance, aircraft manufacturers might use additive manufacturing to create a rapid prototype of a new aircraft turbine wheel designed to maximize air flow. The conceptual rapid prototypes may look like a turbine wheel but built smaller or made of lower-cost material like plastic or foam. A conceptual prototype can help communicate the part design and structural requirements to management, production personnel, and others involved in determining whether or not a part design is feasible for production.

Engineers also use rapid prototyping to create a functional prototype, which are prototypes that match every specification of the part like size, material, and weight. Functional rapid prototyping allows for iterative design, or trying and testing different variations to a part before putting it into full production. In the turbine wheel example, an engineer can use AM to produce multiple fully-functional turbine wheels with slight variations to their design and subject them to rigorous testing. The testing produces data that can help determine which design performs best. Using traditional manufacturing methods, such as CNC machining, would require much larger amounts of the expensive materials used to create aircraft turbines. By combining rapid prototyping methods with iteration, businesses can save costs without sacrificing performance and quality.

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Lesson: 5/9

Design and Prototyping Review

1. Additive manufacturing machines turn a digital model into a physical object by joining or fusing layers of material. **TRUE**
2. Conceptual rapid prototypes may look physically identical to a part but lack its functional qualities. **TRUE**
3. The production stage examines a part prototype to ensure that it performs its designed function. **FALSE**
4. AM prototyping requires machine reconfiguration and retooling. **FALSE**

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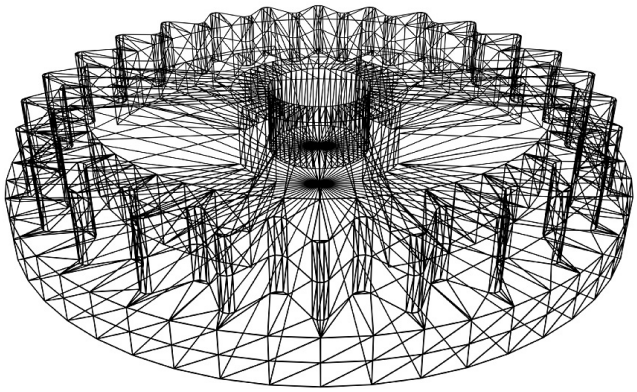
Lesson: 6/9

Rapid Processes for Prototyping

Similar to traditional CNC machining, in the first step of rapid prototyping an engineer creates a part model using CAD software. An engineer typically creates a 3D model of the part because it is more accurate than a traditional two-dimensional (2D) drawing. This generates better outcomes during testing.



After completing the 3D design, the engineer then converts the CAD model to a stereolithography (STL) format. Engineers use STL files for all rapid prototyping processes because of the software's compatibility with AM machines. Converting a CAD file into an STL file converts the 3D model into a sequence of interconnected triangles.

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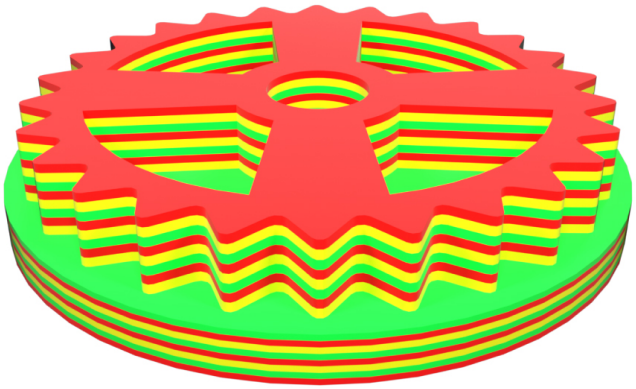
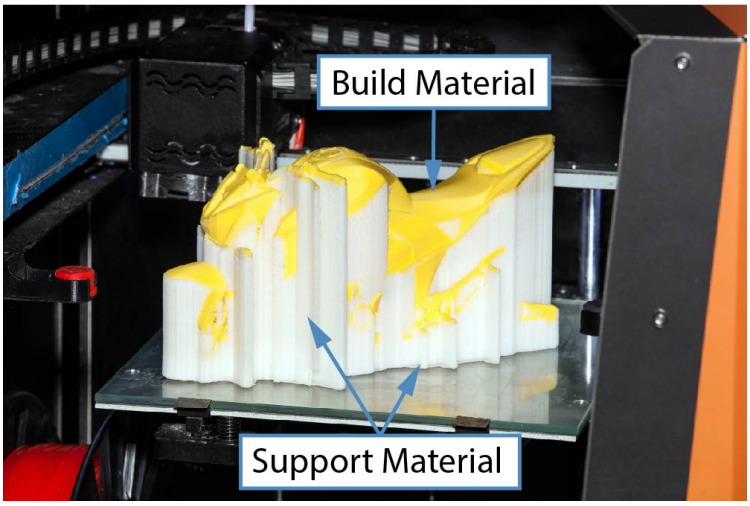


Most AM machines cannot read STL files. Therefore, software must convert the STL file into G Code, a numerical control programming language that a digital slicing program will understand. The digital slicing prepares layers of material that the AM machine will fuse or dispense as it builds the prototype.



The rapid prototyping process, like any additive manufacturing process, is typically automated. Once completed, however, the model may need cleaning or finishing. Also, any support material used in building the model must be manually detached or removed during post-processing.

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Rapid Prototyping Applications

Rapid prototyping has many common applications. The automotive, aerospace, and other transportation industries use rapid processes for lightweighting, in which engineers design vehicle components with the goal of getting the lightest weight possible for the required strength. Lightweighting can involve swapping out traditional materials for lighter ones or creating normally solid parts with internal lattice structures or honeycomb structures. For example, by replacing glass windows with plastic or metal bumpers with carbon fiber, vehicles can achieve better gas mileage without sacrificing safety.

Rapid prototyping allows manufacturers to test part design with a less-expensive material than the actual part would require, which helps reduce costs. In some cases, this can also include rapid prototypes of tooling. For parts made using a mold, engineers can substitute hard tools, or molds made from metals such as aluminum or steel, with soft tools made from less expensive materials such as silicone. For example, using additively manufactured silicone molds for prototyping is more economical than creating metal molds for shorter production runs. Soft tools generally get the same results as hard tools. However, soft tools may only have a quarter of the lifespan of traditional hard tools, so hard tools are more economical for larger production runs. .

Figure 1: Production line for painting carbon fiber bumpers.

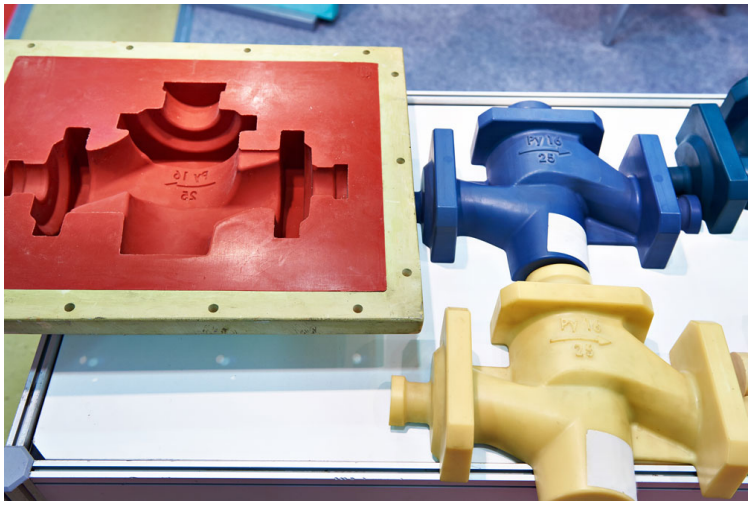

Figure 2: Plastic casting molds.

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Rapid Prototyping: Next Steps

Advances in rapid prototyping depend on the combination of hardware, software, and materials working together. Improvements to this process strive to optimize such things as build times, material preparation, and complexity. For example, newer AM software can work with existing STL files to create more complex lattice structures or innovate with new multimaterial gradients.

Manufacturers may choose to keep prototyping within their organization, or they may outsource to a third party that specializes in rapid prototyping. Advantages to creating prototypes in-house may include quicker turnaround times. Third-party prototyping operations may take over a week, including transporting. However, a physical prototype can potentially be made in-house and consumer-tested all in the same day. In-house prototyping also provides greater security against intellectual property (IP) theft, as digitally transferring CAD or STL files poses cybersecurity risks.

Some disadvantages to in-house rapid prototyping can include significant up-front costs. In-house AM requires purchasing materials and machines, and can also involve the time and cost of training staff and regular machine maintenance. Using a third-party rapid prototyping company avoids up-front costs, but is more expensive if used regularly. However, third party rapid prototyping companies may provide more options for production processes and materials than an in-house operation.

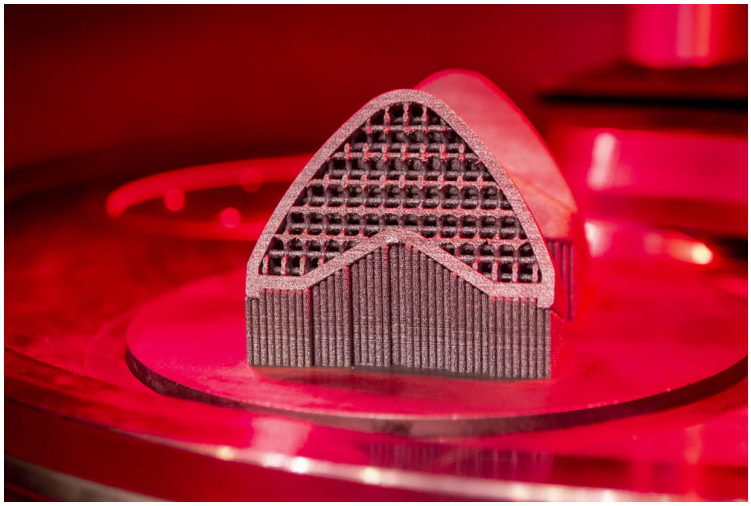

Figure 1: Object made from laser sintering.

Figure 2: Cybersecurity provides greater protection against intellectual property theft.

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Lesson: 9/9

Final Review

Which of the following describes an advantage of in-house rapid prototyping? In-house rapid prototyping:

- **Provides quick turnaround times from part to production.**
- Avoids upfront costs associated with machine purchasing.
- Avoids upfront costs associated with training staff.
- May take a week to complete and transport.

What rapid prototyping products have a short lifespan but can be useful for short production runs?

- **Soft tools**
- Hard tools
- CAD
- STL

What file format is compatible with all rapid prototyping processes?

- **STL file format**
- Setup file format
- Post-processing file format
- CNC file format

Which of the following describes testing the internal structures of a vehicle with the goal of getting the lightest weight for strength?

- **Lightweighting**
- Soft tooling
- Hard tooling
- Stereolithography

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Class Vocabulary

3D Printing The process of producing a 3D object using a specialized tool that creates successive layers of material. 3D printing must technically involve the use of a print head or nozzle, but the term is often used interchangeably with additive manufacturing (AM).

Additive Manufacturing AM. The process of joining or solidifying materials to make an object based on a three-dimensional computer model. Additive manufacturing methods typically build up layers of material to create an object.

AM Additive manufacturing. The process of joining or solidifying materials to make an object based on a three-dimensional computer model. AM methods typically build up layers of material to create an object.

Bar Stock Raw material in the form of long bars or cylinders. Bar stock is commonly used in metal manufacturing processes.

Blueprint A design representing the specific dimensions and production considerations for a part or prototype. Blueprints are created during the design phase of design for manufacturing (DFM).

CAD Computer-aided design. A computer system used to design a model of a product. CAD models are converted into STL format for use by additive manufacturing machines.

Computer Numerical Control CNC. A programmable system of software and hardware that directs the operation of a machine. Computer numerical control systems use mathematical data to direct machine movements.

Computer-Aided Design CAD. A computer system used to design a model of a product. Computer-aided design models are converted into STL format for use by additive manufacturing machines.

Conceptual Models A nonfunctional model used to physically demonstrate design ideas for a part. Conceptual models are often composed of polymer and made using additive manufacturing (AM) processes such as material extrusion or vat photopolymerization.

Conceptualization To form an idea about a possible part or manufactured good. Conceptualization is one of the initial stages of design for manufacturing (DFM) or design for additive manufacturing (DFAM).

Cybersecurity Protection against criminal or unauthorized access to computer networks, programs, and data. Cybersecurity has become a major industrial concern as networking and connectivity have increased.

Designing The process of creating the actual part specifications. Designing involves creating the blueprints and prototypes for a part.

Direct Metal Laser Sintering DMLS. An additive manufacturing process that uses a laser to bond successive layers of material in a bed of powdered metal. Direct metal laser sintering can produce complex metal parts, though they often require extensive post processing.

Finishing The treatment of a surface to remove roughness and irregularities and improve its appearance. Finishing processes include sanding, heat treating, and painting.

Functional Prototype A part model produced through additive manufacturing that represents the composition and design of the final part. Functional prototypes are used to test how a part will perform in its intended application.

Functionality The ability of a part to optimally perform a set purpose. Functionality is a key design consideration.

G Code A method of programming that pairs address letters with numerical values to form words. G code programs are used in additive manufacturing, CNC machining, and hybrid machining.

Hard Tools Higher quality of molds that are usually made of metal. Hard tools have long lifespans suitable for long production runs, but are expensive to manufacture.

Hardware The physical equipment used in a computer system. Hardware includes a computer screen, keys, and power supply.

Honeycomb Structures A type of built structure in which empty cells are separated from each other by a solid wall. Honeycomb cells can be square or hexagonal.

Intellectual Property IP. Any unique idea, creative expression, or knowledge that originated from, or is owned by, a specific person or business. Intellectual property is not physical but has commercial value.

Iterative Design Repeatedly testing a prototype. Iterative design requires testing different variations to a part before putting it into full production.

Lattice Structures A repeating, symmetrical pattern of crossing strips of material that leave diamond- or square-shaped gaps between them. Lattice structures provide excellent strength to a part.

Lightweighting Replacing one part or object with a lighter version for identical use. Lightweighting is a manufacturing concept that lighter parts will lead to greater efficiency.

Machining A subtractive manufacturing process that involves removing material to form an object. Machining includes methods

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such as milling, turning, and drilling that remove metal using cutting tools.

Mill A multi-point cutting tool that is used to remove metal from the surface of a workpiece. Mills are used to cut materials to the near net-shape of a final part.

Mold A hollow cavity that holds heated liquid material and imparts its shape on the material as it cools. Molds with exceptional accuracy can be created using an additively manufactured prototype as the basis for the model.

Multimaterial Gradients The use of multiple materials to create additive manufacturing (AM) objects. Multimaterial gradients allows for greater innovation with AM processes like rapid prototyping.

Outsource To procure products or services from an outside company. Part makers that outsource processes must still meet company requirements.

Planning Assessing the ways to best create the conceptualized part. Planning involves initial considerations of the design and of the practical concerns related to the manufacturing process.

Post-Processing A procedure, or series of procedures, for transforming constructed parts into their final size, shape, and structure. Cutting, drilling, assembly, buffing, and coating are post processing operations.

Production The manufacturing of a finished part for delivery to a customer. Production is the final stage of design for manufacturing (DFM).

Prototype A preliminary model of a part used to evaluate the look and performance of a design. Prototypes are used to determine the specifications for the final part.

Rapid Prototyping A product development technique in which additive manufacturing (AM) methods are used to create prototypes for a traditional manufacturing operation. Rapid prototyping allows engineers to quickly create a number of prototypes in a short time period, reducing lead time.

Raw Materials An unprocessed or lightly-processed object or substance used to make a finished part. Raw materials in manufacturing include metal, plastic, and ceramic.

Setup All of the tasks and preparations necessary to prepare for a machining operation. Setup can include preparing machines as well as collecting paperwork, tools, and materials.

Silicone A polymer that can exist in various forms. Silicones are known for their heat and weather resistance.

Soft Tools Lower quality molds that are usually made of composite materials like silicone or carbon fiber. Soft tools have a short lifespan but can be useful for short production runs.

Stereolithography STL. Computer files that represent 3D models as a series of interconnected triangles. Stereolithography files are converted from CAD files and are used in additive manufacturing machines.

Support Structure Additional material used in the additive manufacturing process to support the product as it is being constructed. Support material is removed from the AM product as part of the additive manufacturing process.

Testing The examination of a part to ensure that it performs its intended function and that it can be satisfactorily manufactured. Testing indicates whether the part needs additional planning or if it is ready for production.

Three-Dimensional 3D. Occurring within three planes of space. Three dimensional objects have length, height, and width.

Topology The arrangement of geometric or material components in a part. The topology of additively manufactured parts can be particularly complex because AM is capable of creating intricate shapes.

Two-Dimensional 2D. Having height and width. Two-dimensional objects are flat and lack depth.

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